Alok Bhatt

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Education

Graphic Era University **August 2019 - May 2023**
Bachelor of Technology in Computer Science and Engineering | CGPA 9.25 *Dehradun, Uttarakhand*

Experience

Arista Networks **Oct 2023 - Present**
Software Engineer *Bangalore*

- Contributed to SDN and OS components of Arista's Multi-Domain Segmentation Service, building an East-West firewall using a high-performance DPDK-based software pipeline (BESS) to enforce Zero Trust policies across intra-data center traffic.
- Designed and implemented firewall policy enforcement logic across control and data planes, including label generation, CLI integration, and support for double-encapsulated tunneled packets in microsegmentation use cases.
- Integrated Qosmos DPI engine into the firewall datapath to enable Layer 7 application visibility, enhancing rule granularity and enabling content-aware policy actions at runtime.
- Engineered features for hitless rule updates, system restart/revert handling, and datapath optimizations using flame graphs and performance profiling tools, achieving a 30% throughput improvement under high-load conditions.

Arista Networks **Jan 2023 – June 2023**
SWE Intern *Bangalore*

- Implemented core dump collection enhancement technique for kernel panic occurrences, utilizing U-Boot side scripting to enable compression logic and inline memory compression for ramdumps.
- Created a user-space application to facilitate the upload of collected dumps to an HTTP server, ensuring efficient and reliable data transfer.
- Acquired in-depth knowledge of kernel drivers, exploring various aspects of wifi technology, including wifi frames and their functioning.
- Modified the pipelines codebase to improve abstraction, resulting in better code maintainability and reduced technical debt.

Amazon ML School **June 2022 – July 2022**
Trainee *Remote*

- Acquired good knowledge in key areas, such as feature engineering, model evaluation, hyperparameter tuning, and optimization algorithms, to solve complex real-world problems.
- Executed a prominent project on customer segmentation using k-means clustering, showcasing the practical application of machine learning to improve decision-making and customer targeting

Google Cloud Sprint Bootcamp **May 2022 – June 2022**
Trainee *Remote*

- 30 days Training from google mentors on Troubleshooting ,Linux and Networking. Developed expertise in troubleshooting complex issues, analyzing system logs, and diagnosing performance bottlenecks within cloud-based environments.
- Designed a scalable Twitter-like platform, with key functionalities including user follow/unfollow and tweet posting. Also created comprehensive documentation for ongoing reference and knowledge sharing.

IEEE **May 2021 – July 2021**
Research Intern *Remote*

- Developed a model for monitoring potential risk areas and detecting forest fires early,such that response time can be significantly reduced and the cost of potential damage and fire fighting can be reduced.
- Conducted extensive data analysis and preprocessing to ensure the model's accuracy and reliability, utilizing data visualization techniques to identify patterns and anomalies in forest data. Achieved a remarkable accuracy of 93% in the model's performance.

Projects

Human body Detection using EM waves | *Python* **October 2022 - Nov 2022**

- Conducted experiments and collected data from controlled environments to validate the correlation between EM wave patterns and specific physiological or emotional states.
- Collaborated with a multidisciplinary team of researchers, including physicists an data scientists, to validate and refine the research findings.

Land Cover Classification using satellite Imagery | *Python* **August 2022 - September 2022**

- Implemented the Random Forest machine learning algorithm for land cover classification, leveraging its ensemble capabilities to enhance classification accuracy and robustness.

Face Mask Detection | *Python* **December 2021 - February 2022**

- Integrated the model with live video streams, allowing for immediate detection of correct face mask usage in real-world scenarios. Achieved 95% accuracy through data augmentation and hyperparameter tuning.

Technical Skills

Core: Data Structures, Algorithms, Object Oriented Programming, Operating System, DBMS, Machine Learning

Languages: C, C++, Python

Tools: Git, Perforce, Docker, Perf, FlameGraph, VIM, VSCode

Cloud & Infrastructure Technologies: DPDK, GCP, Kafka, Jenkins, Cluster Technologies, SQL, MySQL, gRPC, Microservices